

University of Auckland - EE310

Modern Lighting Design Assignment

Team 14

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This project was undertaken on behalf of Alpha Design Limited, with the lighting design and analysis carried out by a student team from the University of Auckland. We also extend our gratitude to our course instructor, Dr Dariusz Kacprzak, our teaching assistants and all other staff who assisted us during this project.

Executive Summary

This report covers the evaluation and redesign of the lighting systems for two existing office buildings, Building A and Building B, purchased by Alpha Design Limited. It also includes the lighting design for a new single-level covered carpark on the same site. The goal of this project was to evaluate the existing lighting against AS/NZS 1680, redesign the lighting to meet compliance, and achieve a payback period of under 10 years.

The evaluation of the existing fluorescent lighting found widespread failures across both buildings. The most common issues were poor uniformity in offices, meeting rooms, CAD rooms, and corridors, along with UGR failures in CAD rooms and several rooms on both floors of Building A and first floor of Building B. The existing system has a total connected load of 25,428 W, with a power density above the recommended 7-9 W/m² range, and produces around 20.98 tonnes of CO₂ per year.

A full lighting redesign was completed using LED luminaires from Lena Lighting, with emergency lighting from Legrand. The new design meets AS/NZS 1680 across all rooms in both buildings, achieving the required illuminance and uniformity for each space, along with the required UGR for CAD rooms. The new system brings the total connected load down to 6,593.9 W, a 74% reduction in energy consumption, reducing annual CO₂ emissions from 20,978 kg to 5,440 kg, saving 15,538 kg CO₂ per year.

Based on a luminaire cost of \$300 per unit and an electricity rate of \$0.25/kWh, the total capital investment is \$83,400, with annual running costs of \$5,440. Emergency lighting has been designed to meet Clauses F6 and F8 of the New Zealand Building Code across both buildings and the carpark.

The design also considered broader responsibilities beyond technical compliance. The 74% reduction in energy consumption and significant decrease in CO₂ emissions supports the United Nations Sustainable Development Goals of SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production) through responsible lamp recycling and selection of long-life LED luminaires, and SDG 13 (Climate Action). The principles of Te Tiriti o Waitangi were also upheld throughout the project, with Partnership reflected in collaboration with Legrand New Zealand and Alpha Design Limited, Participation through the design of inclusive and safely lit spaces for all building users, and Protection through the reduction of environmental impact for future generations.

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1.0 Introduction

1.1 Project Background

This project was completed in collaboration with Legrand New Zealand for Alpha Design Limited, a company that designs packages and boxes. Alpha Design Limited recently purchased two existing two-level office buildings, Building A and Building B, on the same site. A design engineering team was brought in to evaluate the existing lighting and deliver a new lighting design across both buildings. A new single-level covered carpark is also being built on the same site, which needs a full lighting and emergency lighting design.

Both buildings contain offices, meeting rooms, CAD rooms, a kitchen, corridors, toilets, and storage areas across two floors. The existing lighting in both buildings is made up of fluorescent luminaires that were installed before Alpha Design Limited took ownership.

1.2 Project Scope and Deliverables

This project covers the following:

- Evaluation of the existing lighting in Buildings A and B against AS/NZS 1680
- Redesign of the interior lighting for both buildings to meet AS/NZS 1680
- Emergency lighting design for both buildings per clauses F6 and F8 of the New Zealand Building Code
- Lighting and emergency lighting design for the new covered carpark
- A cost plan showing a payback period of under 10 years
- A report on annual CO₂ emissions for both the existing and new systems

1.3 Design Methodology and Tools

DIALux Evo was used to model all rooms across both buildings and the carpark. Each room was simulated with luminaires laid out to meet the illuminance, uniformity, and UGR requirements of AS/NZS 1680, along with emergency lighting according to F6/F8 sections of NZ building code.

Surface reflectances were set as per the project design parameters - ceiling 70%, walls 50%, floor 20%, and windows 10%. Illuminance targets, uniformity requirements, and UGR limits were taken from AS/NZS 1680.2.1 Table D1, AS/NZS 1680.2.2 Table E1, and AS/NZS 1680.1 Table 3.2. Light loss factors were calculated using manufacturer data where available, with AS/NZS 1680.4 used as a reference otherwise.

All luminaires for the new design are from ETAP, Legrand, and Lena Lighting, as recommended by the project brief.

2.0 Evaluation of existing lighting (Building A and B)

2.1 Methodology

The existing lighting was evaluated using DIALux Evo as described in Section 1.3. Working planes were set for each room to get the average illuminance and uniformity. Calculation surfaces were also added to get UGR values for rooms where glare is a concern, such as CAD rooms. Results were compared against AS/NZS 1680.

Working plane heights were set to 0.7m for all occupied rooms including offices, meeting rooms, CAD rooms, the kitchen, and the director's office. Rooms where the working plane is at floor level, such as corridors, stairs, toilets, and DB rooms, were set to 0m.

Full results for both buildings are in Appendix A.

2.2 Compliance with AS/NZS 1680

The existing lighting in both buildings uses fluorescent Troffer 4×36W and Batten 2×36W luminaires supplied by Philips, installed prior to Alpha Design Limited's purchase of the buildings. As shown in Appendix A, the majority of rooms across both buildings meet the minimum illuminance and uniformity requirements of AS/NZS 1680. However, several rooms have notable failures and many rooms that pass on illuminance record lux levels far in excess of what is required, indicating an inefficient installation rather than a well-designed system.

The most consistent failures across both buildings are in corridors, CAD rooms and DB rooms. Corridor uniformity is particularly poor, with Building B first floor Room 202 recording a critically low U_o of 0.049 and Building B ground floor Room 102 recording 0.25, both well below the ≥ 0.3 minimum for circulation spaces. Building A ground floor Room 108 Corridor similarly fails at 0.22. CAD rooms across both buildings fail UGR requirements, with values ranging from 19.4 to 22 against the ≤ 19 limit, indicating glare discomfort for occupants performing prolonged screen-based work. Both buildings also have DB rooms recording zero or insufficient illuminance, suggesting non-functional or absent fittings.

Beyond these specific failures, the existing system consistently over-illuminates many spaces. Building B ground floor CAD room records 1107 lx against a requirement of only 320 lx, and several offices and reception areas exceed their targets by a factor of three or more. This demonstrates that while basic illuminance compliance is largely achieved, the existing fluorescent system is significantly energy inefficient and lacks the uniformity and glare control required for a modern workplace, justifying a full redesign using LED luminaires.

2.3 Power and Power Density Analysis

The total connected load of the existing system across both buildings is 25,428 W. Based on a total building area of 1,173 m², this gives a power density of 21.68 W/m², with full calculations being available in Appendix D3. The recommended limit for office buildings is 7-9 W/m² per NZS 4243.2, which the existing system exceeds in many rooms due to the high wattage of the fluorescent fittings. This is another reason the system needs to be replaced, as the new LED design will bring power density within the recommended range.

2.4 Lifetime of Luminaires and Recycling Costs

The existing system uses fluorescent lamps, which typically last between 24,000 and 30,000 hours (U.S. Department of Energy, 2010) under normal use. LED luminaires by comparison last up to 50,000 hours (Gagnon, 2024) , meaning fluorescent lamps need replacing more often, adding to maintenance and labour costs over time.

Fluorescent lamps contain mercury, classifying them as hazardous waste under New Zealand regulations, which incurs additional disposal costs beyond standard lamp replacement (Interwaste, n.d.). Given the recycling cost of \$10 per T8 lamp, with a total of 678 T8 lamps the total recycling cost is \$6780. This further supports the case for transitioning to LED lighting.

Moving to LED removes these disposal costs, reduces how often lamps need replacing, and lowers the overall running costs for Alpha Design Limited.

2.5 Annual CO₂ Emissions of Existing System

The annual CO₂ emissions of the existing system were calculated using the total connected load, operating hours, and an emissions factor of 0.25 kg CO₂/kWh as set out in the project brief. Office lighting runs for 12 hours a day, 5.5 days a week, 50 weeks a year, 3,300 hours annually. Carpark lighting runs 24 hours a day, 7 days a week, 52 weeks a year.

The existing system produces approximately 20.98 tonnes of CO₂ per year. Full calculations can be found in Appendix D6. This will be significantly reduced with the new LED design, as covered in Section 3.

3.0 Design of a new lighting system (compliant with AS/NZS 1680)

3.1 Building A

We have designed a new lighting system for building A that is designed to comply with AS/NZS standards while reducing power consumption. Notable standards that we complied with were the average luminance, uniformity and unified glare rating (UGR). In Appendix B we compare the values of the AS/NZS 1680.1 and AS/NZS 1680.2.2 standards for luminance, uniformity and unified glare rating with our new system showing our compliance with the standards. The reflected ceiling plan in Appendix G describes the layout of the Luminaires and how they align with the ceiling for installation.

Appendix F1 contains our luminaire schedule for building A. This shows a detailed list of all the Luminaires used in building A with their corresponding amount, power rating, illuminance and luminous efficacy.

Ground Floor:

On the ground floor, the luminaires are predominantly recessed and are fitted into the existing suspended ceiling with a ceiling modulation of 1.2 m by 0.6 m. The lighting system was designed to comply with the required average maintained illuminance, uniformity and

UGR values while reducing power consumption compared to the original fluorescent installation. A 4000 K colour temperature was used to provide a neutral white lighting which is mainly suitable for the working room. This improves the visual clarity in offices, meeting and CAD rooms while remaining appropriate for circulation and general-purpose spaces.

The ground floor contains a mixture of different rooms and requires different lighting requirements, and so several ETAP luminaire types were used rather than applying a single fitting throughout the entire floor.

The U7C-M112-35-840-020D luminaire shown in Figure A was selected for the main circulation spaces. 5 fittings were installed in Room 102 Corridor/Lobby shown in Figure B and 8 were installed in 108 Corridor/Lobby. Each luminaire produces 2,097 lm. The linear shape and broad light distribution make this fitting suitable for long circulation routes as it helps distribute light consistently throughout. This reduces dark patches and improves visibility for occupants moving through the building. As corridors are used primarily for safe movement rather than desk based work, a UGR less than 25 was considered appropriate.



Figure A: U7C-M112-35-840-020D

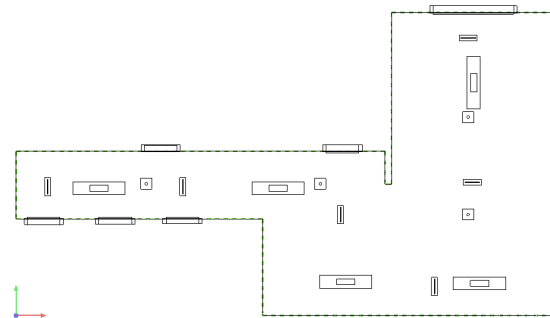


Figure B: 102 Corridor/Lobby Room

Room 101 stairs shown in Figure C required a compact fitting with sufficient output to illuminate the stair treads and landings clearly. One surface mounted R820R1/LEDN25D shown in Figure D was used, producing 2499 lm. The fitting was selected to provide an evenly distributed light throughout the stairs. This improves safety by reducing the abrupt changes in brightness.

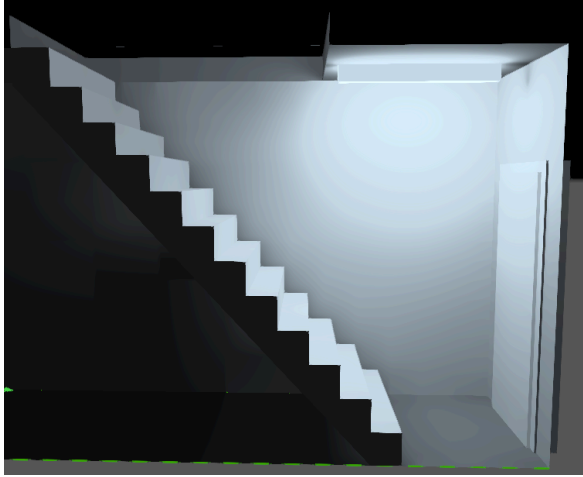


Figure C: 101 Stairs



Figure D: R820R1/LEDN25D

The U2B recessed luminaire family was used extensively in meeting rooms, offices, mail and CAD rooms. These luminaires were arranged neatly into the suspended ceiling and provide a diffused light output, making them suitable for spaces where occupants stay seated for a long time. The diffusion helps reduce visual discomfort while distributing lights evenly through the working plane.

The meeting rooms and offices as seen in Figure E each used U2B-M210-MI-840-025D luminaires shown in Figure F. Each fitting produces 2495 lm. The positioning of these lights on the ceiling was designed to provide a consistent spread of light over objects such as meeting tables etc. These arrangements support visual comfort during meetings and screen-based presentations while avoiding unnecessary over-lighting.

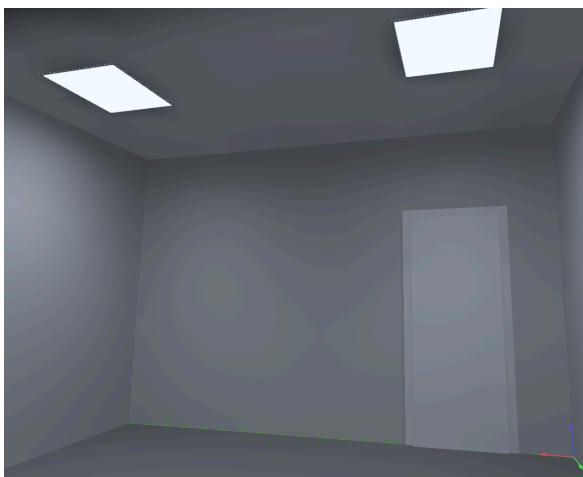


Figure E: 110 Office Room

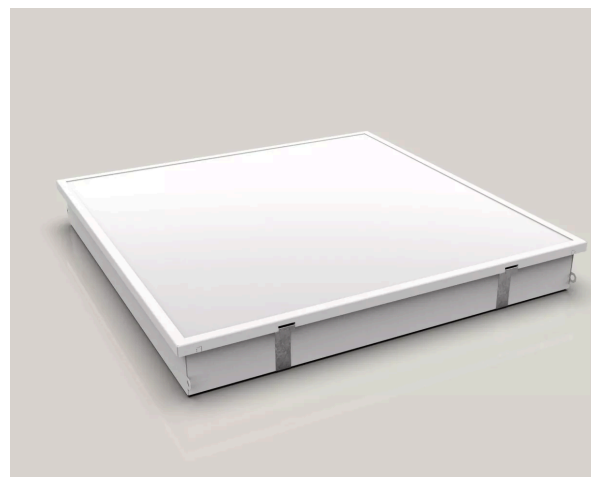


Figure F: U2B-M210-MI-840-025D

All luminaires selected for Building A maintain a CRI of $R_a \geq 80$, meeting the minimum Colour Rendering Group 2 requirement of AS/NZS 1680 Table D1. However, standard LED luminaires frequently exhibit poor R9 values, a measure of how accurately saturated red

tones are rendered despite meeting the overall CRI ≥ 80 threshold. In CAD rooms where colour-coded engineering drawings and red dimension lines are regularly used, luminaires with enhanced R9 performance were prioritised during selection. This improves the accurate perception of red tones, reducing the risk of colour misidentification during detailed design work and enhancing occupant comfort and safety.

The 2 CAD rooms required particular attention because occupants complete prolonged screen-based tasks. Excessive glare or uneven illumination can reduce visual comfort which can make working harder. Room 111 CAD uses nine U2B-M210-MI-840-025D luminaires while Room 112 CAD uses six higher-output U2B-M210-MI-840-045S luminaires U2B-M210-MI-840-045S each provide 4490 lm. This higher output was selected to achieve the required luminance without increasing the number of fittings. In both CAD rooms, the luminaires were arranged to improve uniformity across the working plane and reduce reflections on computer screens. The final layout ensured the UGR remained ≤ 19 .

Room 109 Mail shown in Figure G uses 3 U2B-M110-MI-840-025D. The recessed fitting was appropriate because the room required clear general illumination for sorting out and handling mail. The placements were designed to be spread across the task area while maintaining a low connected load of 51 W.

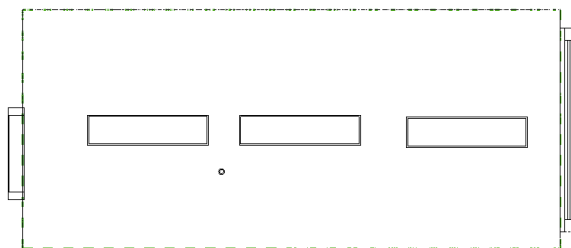


Figure G: 109 Mail Room

Smaller spaces such as the toilets shown in Figure H were designed with compact luminaires. All toilet rooms use 1 of D9B-R310-10-840-015D luminaires shown in Figure I. This produces 1450 lm. The compact fitting provides general lighting across the room. Toilet areas also use luminaires with an ingress protection rating suitable for moisture-prone environments.

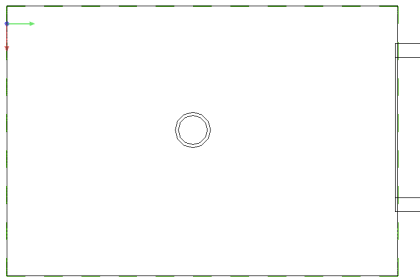


Figure H: 103 Toilet Room



Figure I: D9B-R310-10-840-015D

Room 114 Visitors used 2 D1C-R300-11-840-015D luminaires. Each produces 1447 lm. This was suitable for smaller visitor areas because they provide sufficient illumination while keeping the total connected load low. In the reception area and visitor room, luminaires were deliberately positioned to enhance vertical surface illumination. Vertical illuminance on walls and vertical surfaces improves the perception of faces and objects, which is particularly important in reception and visitor-facing spaces where occupants interact directly. This supports visual comfort and creates a welcoming first impression for visitors to Alpha Design Limited.

First Floor:

On the first floor similar design decisions were made to follow compliance with AS/NZS. A notable difference in the first floor is the Luminaires being surface mounted compared to suspended like the ground floors. The selected luminaires were used to help get a unified glare rating of less than 19 and keep the room bright with sufficient uniformity. Another difference is the balcony rooms which are open and expose the lights to the outdoor environment. To account for this an IP66 luminary E2A-S102-1P-840-030D was used to protect from moisture during heavy rain and storm and potential solid objects. Like the ground floor 4000K colour lamps were chosen with the same goal of using cooler lighting to improve focus and concentration. On the first floor we used two Luminaires, one with a square housing at 21W and a linear housing at 25W. Both Luminaires are selected and installed in specific locations to help achieve certain goals such as uniformity, glare and luminous efficacy. Full compliance of AS/NZS standards for this floor is detailed in Appendix B.

The most common luminary used on the first floor is the COMPACT LED EVO N 3100LM shown in Figure J. This luminary is excellent in its versatility and performance. The luminary has a 4000k lamp colour with a CRI of $R_a \geq 80$ which provides a cooler lighting for productivity and focus while keeping accurate colour rendering. The Luminaires are installed in corridors, offices, manager and meeting rooms. At 25W the luminaire has a luminous flux of 3100lm it was suitable for the bigger and more open rooms and the linear design allowed the rooms to meet illuminance, uniformity and glare compliances. Notably this was also used in the CAD room shown in Figure K which requires a high luminance and uniformity while maintaining a unified glare rating of less than 19 which was achieved by using this luminary.

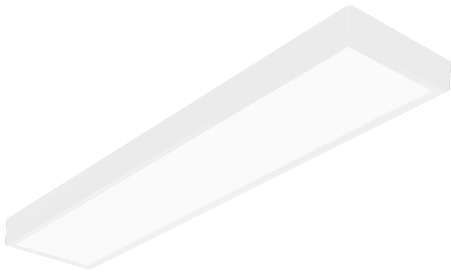


Figure J: COMPACT LED EVO N 3100LM



Figure K: 210 CAD Room

The ARTO 2 LED 2250LM shown in Figure L was also selected on similar design decisions as the COMPACT LED due its versatility and performance in various rooms. The ARTO LED was also used in various rooms like corridors, offices and meetings due to having similar performance as the COMPACT LED. The luminary has 4000k lamp colour to create an environment that is suitable for tasks at a colour rendering index of $Ra \geq 80$. The ARTO 2 LED only uses 21W and so has a lower luminous flux of 2250lm. This was suitable for smaller rooms or rooms that are sufficient at lower illuminance's like toilets, stairs and common rooms shown in Figure M. The square housing provides rooms with high uniformity and maintains a unified glare rating of less than 19 for task rooms.

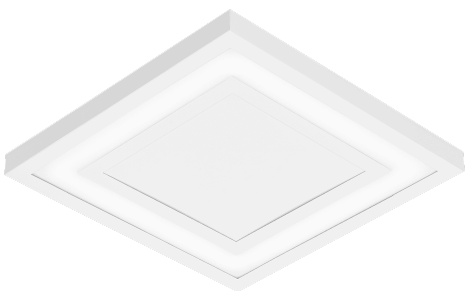


Figure L: ARTO 2 LED 2250LM



Figure M: 212 Common Room

For balconies, the rooms are exposed to outdoor environments and so E2A-S102-1P-840-030D shown in Figure N has been chosen as a suitable luminary for these conditions. Since balcony rooms shown in Figure O are open, this means the room and therefore luminary is susceptible to higher moisture levels and water damage due to potential storms and heavy rain. The E2A luminary provides an IP rating of IP66 which has high protection against solid objects and moisture making it sufficient for the balconies.



Figure N: E2A-S102-1P-840-030D

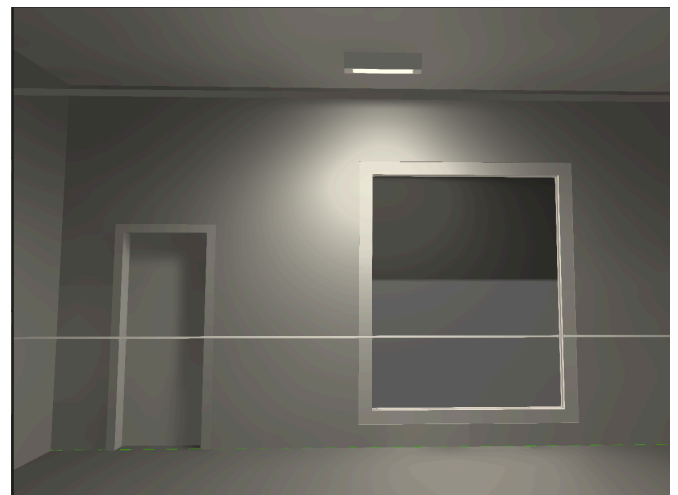


Figure O: 205 Balcony Room

3.2 Building B

Building B was redesigned using similar principles to Building A, with the aim of achieving compliance with AS/NZS 1680 while improving the visual comfort and reducing power consumption. The existing fluorescent system provided excessive illuminance in several rooms but did not distribute the light evenly. Therefore, the redesign focused on improving uniformity through more appropriate lighting spacing and selecting LED fittings with light distribution suited to the dimensions of each space. The reflected ceiling plans for both floors are shown in Appendix G, while the compliance results for illuminance, uniformity and UGR are presented in Appendix B. A 4000K colour temperature was maintained throughout for a neutral professional working environment, with all luminaires achieving CRI Ra $\geq$ 80.

Ground Floor:

On the ground floor we have designed a new lighting system that fits into the existing ceiling modulation of 1.2m by 0.6m and complies with AS/NZS standards. Design decisions were made to also improve comfort and the power efficiency of the Luminaires. Appendix B demonstrates how compliance standards have been met for luminance, glare and uniformity.

The new reflected ceiling plan is shown in Appendix G. The reflected ceiling plan shows how the Luminaires fit the 1.2m by 0.6m ceiling modulation.

The floor contains a mixture of circulation areas, offices, meeting rooms, storage spaces, toilets, a shower, reception area and a CAD room. As these spaces have different lighting

requirements, a combination of linear and square LED luminaires was used rather than applying a single fitting across the entire floor.

The main luminaires used on the ground floor were the COMPACT LED EVO N 3100 lm, COMPACT LED EVO P ZK 3550 lm and ARTO 2 LED fittings. Three ARTO 2 models were selected with outputs of 2250 lm, 3950 lm and 5270 lm. All fittings have a colour temperature of 4000 K, which provides a neutral- white light suitable for working environments. The cooler appearance was selected to improve visibility and support concentration in frequently occupied spaces such as offices, meeting rooms and the CAD room.

The linear COMPACT LED EVO N luminaire was used extensively throughout the floor. Its elongated shape provides a wider spread of light than a concentrated fitting, making it suitable for corridors, stairs and larger occupied rooms. For example it was used in Room 102 corridor shown in Figure P and 117 corridor, Figure Q to improve the distribution of light along the corridor and reduce dark areas while fitting.

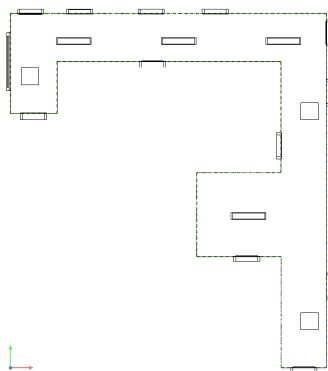


Figure P: 102 Corridor Room

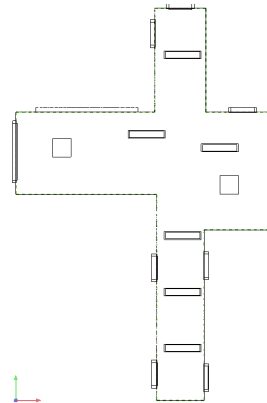


Figure Q: 117 Corridor Room

The same linear luminaire was also used in Room 110 CAD shown in Figure R and Figure S. The existing CAD room produced a high illuminance level but had poor uniformity. 10 COMPACT LED EVO N were arranged to provide a more consistent spread of light throughout the room. CAD work often prolongs longer screen usages, glare control was especially important. The final design had to meet the UGR limit of less than or equal to 19 while maintaining illuminance and uniformity.



Figure R: 110 CAD Room

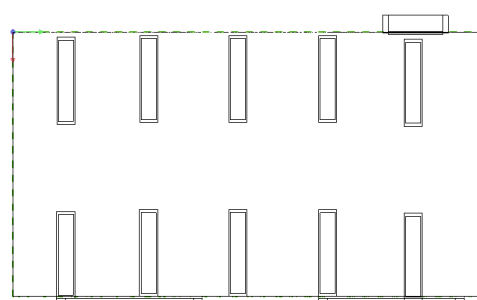


Figure S: 110 CAD Room

The recessed ARTO 2 LED fittings were selected for rooms where square Luminaires were better suited to the ceiling grid and room proportions. Different lumen outputs were used depending on the size and requirements of each space. Lower out-put 2250 lm fittings were used where a softer contribution was sufficient, while the 3950 lm and 5270 lm models were applied selectively in rooms requiring additional output.

Some rooms use a combination of linear and square fittings to improve the overall distribution of light. For example, Room 108 Office shown in Figure T uses 6 COMPACT LED EVO N luminaires and four ARTO 2 LED 2250 lm luminaires. Room 115 reception shown in Figure U, Room 117 Corridor and Room 120 Office can also use mixed layouts. Combining Luminaires allowed the arrangement to be adapted to each room rather than apply a single repeated grid pattern.

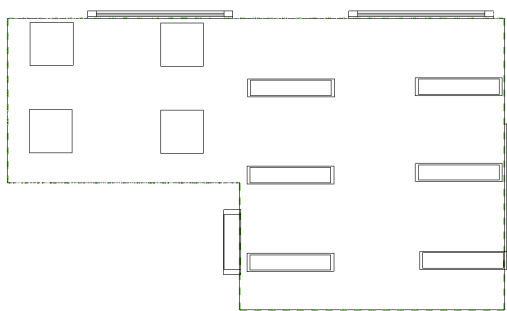


Figure T: 108 Office Room

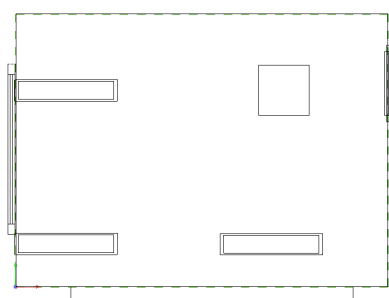


Figure U: 115 Reception Room

The toilets, shower and storage spaces were considered separately because they require general illumination rather than prolonged task lighting. The COMPACT LED EVO P ZK fitting was used in Rooms 103 storage, 105 WC, 106 WC and 107 shower. A wide distribution was appropriate in these rooms to illuminate the entire space evenly. For moisture-prone locations such as showers and toilets, the final fitting selection must also provide a suitable ingress protection rating for the environment.

Also, Room 113 Corridor/Coffee was recognised as a breakup space where occupants step away from desk-based work, and was treated differently from the surrounding task areas accordingly. A warmer colour temperature of 3000K was selected for this space to create a more relaxed atmosphere distinct from the 4000K used throughout the rest of the floor, supporting mental recovery during breaks. This deliberate shift in lighting character helps define the space as a social and rest area without requiring any physical partitioning.

First Floor:

The first floor of Building B features a 3.3m exposed concrete slab ceiling throughout, which prevents the use of recessed luminaires. As a result, all luminaires are either surface mounted or suspended. Where suspended luminaires were used, a 500mm drop from the ceiling was applied to reduce the cave effect, improve ceiling luminance and enhance visual comfort for occupants. A 4000K colour temperature was maintained throughout the floor for consistency with the rest of the building, with all luminaires achieving a CRI of $R_a \geq 80$ to ensure accurate colour rendering across all spaces. The exception is the kitchen which uses

3000K to provide a warmer, more relaxed atmosphere appropriate for a breakout space. The maintenance factor applied to all rooms was calculated in Appendix D5 using the L70B50 lifespan data for each luminaire type. Full compliance results for average maintained illuminance, uniformity and UGR are presented in Appendix B, and the luminaire schedule is detailed in Appendix F.

Circulation spaces use the Baris 40 LED N VD (Lena Lighting), shown in figure V, a fitting specifically designed for traffic zones and circulation spaces per AS/NZS 1680.2.1. In the stair room a single suspended unit was positioned toward the upper corner, as shown in figure W, to minimise glare for ascending occupants and improve tread and riser contrast for safe use. The opal diffuser was selected to reduce LED source visibility and provide soft, diffuse output. The corridor uses the same fitting family at higher output to maintain the 10:1 illuminance ratio required between adjacent spaces, while the wide sideways distribution provides adequate wall illuminance for people and object recognition.

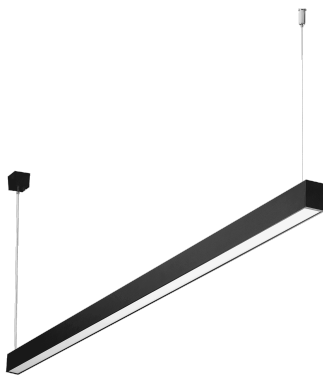


Figure V: Baris 40 LED VD

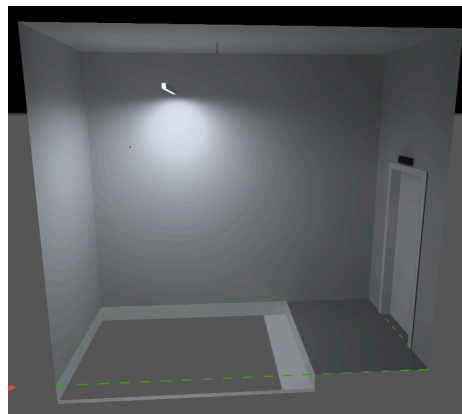


Figure W: 201 Stairs

The Terra 2 LED ZW 595×595mm Multiwatt (Lena Lighting, 24-36W, 4000K, $R_a \geq 80$), shown in figure X, was selected as the primary fitting for the director's room, offices, meeting rooms, CAD room and conference room, all suspended 500mm below the concrete ceiling to an effective height of 2.8m. The client's requirements prioritised average maintained illuminance and uniformity as the primary compliance criteria. Beyond this requirement, the shielded louvre optic of the Terra 2 achieves $UGR \leq 19$ across the majority of task rooms as an additional design improvement, benefiting meeting presentations, screen-based work and general visual comfort throughout the floor. The wide square format provides even illuminance across the working plane with good wall illuminance for any shelving, noticeboards or presentation materials. The multiwatt feature allows output to be tuned to each room size without changing fitting type, maintaining a consistent luminaire schedule across the floor.



Figure X: Terra 2 LED ZW



Figure Y: 203 Director

Notable challenges and exceptions are as follows. Meeting room 208, shown in figure Z, incorporates two additional Terra 2 LED Long Z (1195×230mm, 36W, 4000K, black) fittings to improve uniformity and provide wall illuminance along the presentation wall in accordance with AS/NZS 1680.2.2. Office 205's L-shaped layout caused uniformity of cold spots in the alcove area, resolved by adding a surface mounted Baris 40 LED UGR Plus N (1140mm, 19W, 4000K, 70D wide beam) in the bend. The 70D beam angle was selected over the narrower 60D option to maximise light spread across the confined alcove area, improving uniformity. Conference room 213, shown in figure 1A, similarly required two supplementary Baris 40 VD surface mounted fittings to achieve the required $U_o \geq 0.5$, with surface mounting reducing UGR from 20 to 18.7.

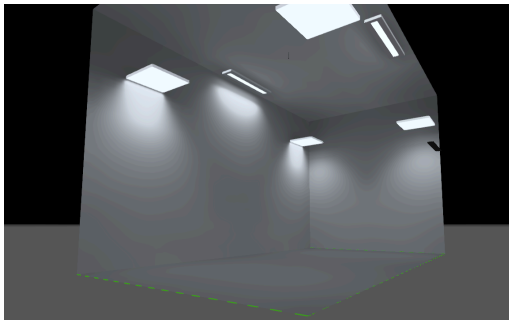


Figure Z: 208 Meeting

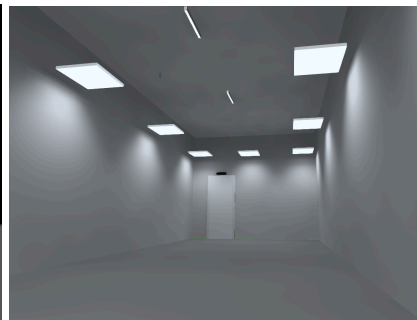


Figure 1A: 213 Conference

Wet areas including the kitchen, shown in figure 1B, and toilets use the Compact LED EVO P ZK (Lena Lighting, IP44, $R_a \geq 80$) for moisture protection in accordance with the ingress protection requirements for wet environments. The kitchen uses the 830 (3000K) variant, surface mounted for ease of cleaning and to reduce the risk of food splatter on the fitting body. The toilets use the 840 (4000K) variant for a clean, clinical appearance appropriate for a workplace bathroom.



Figure 1B: 206 Kitchen

4.0 10-year payback time

4.1 Payback Time

In the context of the redesign of buildings A and B, payback time is dependent on two factors; the number of new luminaires, and the power used by these new luminaires. It would thus follow that minimising these two parameters whilst still satisfying the minimum standard compliance requirements in simulation would yield the best results.

Therefore in order to reduce the client's payback time the new lighting systems have been designed in accordance with these principles; the total number of Luminaires has been minimized wherever possible, low-power luminaires were implemented as much as possible, and through design simulations and compliance checks the efficiency and standards adherence of the system has been ensured.

4.2 Assumptions and Parameters

The calculations for payback time are dependent on a few assumptions; the Alpha Design office hours are fixed at 3300 hours per year, any costs related to emergency lighting or carpark lighting are to be excluded, every new LED luminaire has a fixed cost of \$300, and finally that net energy price is constant at \$0.25/kWh.

4.3 Accumulated Cost Graph

The payback time is shown below in Figure 1C. Payback is achieved around year 6, which is well below the 10 year requirement.

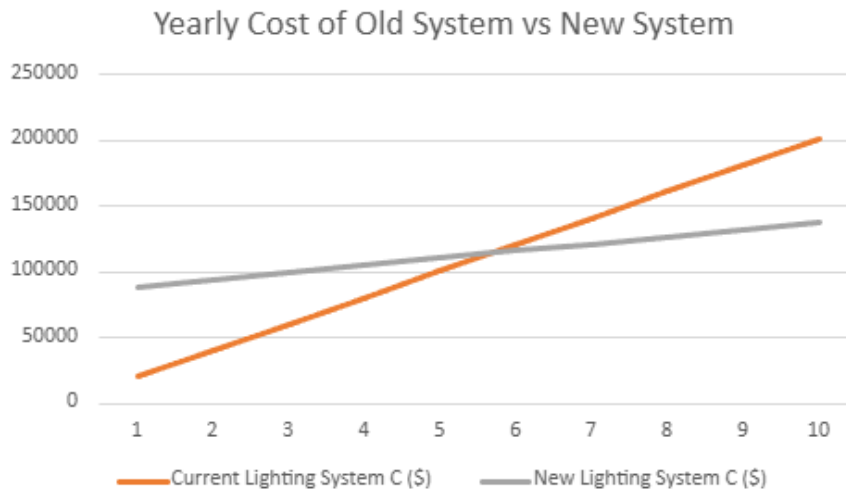


Figure 1C: Annual cost of existing system versus proposed system to find payback time

4.4 Packback Time Calculation

Refer to Appendix D4 for supporting calculations.

5.0 Redesign of lighting in carpark

5.1 Reflected Ceiling Plan

The reflected ceiling plan for the carpark is presented in Appendix G. The carpark is a single-level covered structure with around 2000m² of area, housing two car entrance/exit zones, 3 human exit doors, a generator room, and a storage room. The carpark had no existing lighting installation prior to this project, and a complete new lighting system was designed and commissioned from scratch to meet the requirements of AS/NZS 1680. Luminaires were positioned above driving aisles to maximise floor-level illuminance and minimise shadows within parking bays. The positioning of luminaires was carefully planned to account for structural columns within the carpark, with fittings offset from column locations to prevent shadowing on driving surfaces. The entrance zones were treated as separate lighting zones to meet the significantly higher illuminance requirements specified in AS/NZS 1680 Table D1.

5.2 Compliance with AS/NZS 1680 Standards

The carpark lighting design was evaluated in DIALux Evo using photometric files provided by ETAP. The working plane was set to 0.0m for all carpark zones, reflecting floor-level illuminance as the relevant measurement surface for vehicle and pedestrian movement. Surface reflectances of 70% ceiling, 50% walls, and 20% floor were adopted in accordance with AS/NZS 1680.1 recommended values for interior lighting design. The maintenance factor was set to 0.83 for the carpark and generator room, and 0.86 for the storage room, calculated in accordance with AS/NZS 1680.4 as detailed in Appendix D5.

Both entrance zones spanning the first 15m from each entry point maintained illuminance levels of 815 lux and 816 lux respectively, satisfying the 800 lux daytime target specified in AS/NZS 1680 Table D1 clause 11.1(a). Both zones achieved uniformity values of 0.54 and

0.51, well above the 0.25 minimum, ensuring consistent and even illumination for safe driver eye adaptation from external daylight conditions. The transition zones spanning the subsequent 4m achieved maintained illuminance levels of 175 lux and 222 lux respectively, comfortably exceeding the 160 lux requirement of clause 11.1(a), with uniformity values of 0.38 providing a smooth visual transition into the general carpark interior.

The illuminance transition from the 800 lux entrance zone to the 160 lux transition zone represents a ratio of 5:1, and from the 160 lux transition zone to the 40 lux general parking area represents a ratio of 4:1. Both transitions remain within the maximum 10:1 illuminance ratio between adjacent spaces recommended by AS/NZS 1680.1, ensuring comfortable visual adaptation for drivers moving progressively through the carpark environment.

The general parking and aisle area achieved a maintained illuminance of 110 lux, significantly exceeding the 40 lux minimum required by AS/NZS 1680 Table D1 clauses 11.3 and 11.4, with a uniformity of 0.26 satisfying the 0.25 minimum requirement. This confirms that consistent and safe visibility is maintained throughout the driving surfaces and parking bays for both vehicles and pedestrians.

The storage room achieved a maintained illuminance of 118 lux, exceeding the 80 lux live storage requirement of AS/NZS 1680 Table D1 clause 8.1(b), with a uniformity of 0.69 surpassing the 0.50 minimum. The UGR of 23 satisfies the maximum glare limit of 25 specified for live storage areas per Table D1 clause 8.1(b). The generator room achieved a maintained illuminance of 114 lux, exceeding the 80 lux general plant room requirement of AS/NZS 1680 Table D1 clause 9.1, with a uniformity of 0.51 surpassing the 0.50 minimum. The UGR of 24 satisfies the maximum glare limit of 25 per clause 9.1. The consistently strong performance across all carpark zones confirms that the selected luminaires and placement strategy deliver a reliable and compliant lighting system throughout the entire carpark facility. The full numerical results and compliance summary are presented in Appendix B.

5.3 Key Design Decisions

Several key design decisions were made to ensure the carpark lighting system meets both technical standards and practical operational requirements.

New Installation from Scratch

The carpark had no prior lighting system in place. All luminaires, wiring, and control systems were designed and specified as a complete new installation. This provided full freedom in luminaire selection, placement, and control strategy, allowing the design to be fully optimised for compliance with AS/NZS 1680 without the constraints of an existing infrastructure.

LED Technology Selection

LED technology was selected throughout the carpark in place of traditional fluorescent or HID sources. LED luminaires offer significantly longer operational lifespans of up to 50,000 hours, lower energy consumption, and instant full-brightness operation particularly important in a carpark environment where lights operate 24 hours per day, 7 days per week, totalling 8,736 operating hours annually. This continuous operation makes the high efficiency and

long lifespan of LED technology essential for minimising both energy costs and maintenance requirements over the life of the installation.

Fitting Selection and IP Rating

The ETAP E62/LED2N30S was selected as the primary luminaire for the general carpark area and generator room. This fitting features an IP66 rating, which exceeds the minimum requirement for enclosed carpark environments subject to vehicle exhaust gases, dust, and moisture. The E6 series was specifically developed for parking garage applications and provides inherent resistance to exhaust gas ingress, making it particularly suited to this environment in accordance with AS/NZS 3000 requirements for harsh indoor environments. Reusing the same fitting across the general carpark and generator room also simplifies procurement and future maintenance by reducing the number of distinct product types on site.

For the entrance zones, the ETAP E62/LED4N60D was selected due to its higher lumen output of 6,000 lm and DALI dimming capability, which is essential for meeting both daytime and nighttime illuminance requirements as discussed below.

For the storage room, the ETAP E5B-S714-2A-840-050S was selected. This fitting carries an IP20 rating, appropriate for the dry enclosed indoor environment of a storage room per AS/NZS 3000, and provides 5,000 lm output sufficient to meet the 80 lux live storage requirement of AS/NZS 1680 Table D1 clause 8.1(b).

CCT and CRI Selection

A correlated colour temperature of 4000K was selected for all carpark luminaires. Neutral white light at 4000K promotes alertness and provides good colour contrast, enhancing driver and pedestrian visibility and safety within the carpark environment. A minimum CRI of 80 (Colour Rendering Group 2) was maintained across all zones in accordance with AS/NZS 1680 Table D1, which specifies lamp colour rendering Group 2 or 3 as a minimum for all carpark lighting applications.

Luminaire Placement Strategy

All luminaires were positioned above driving aisles rather than directly above parking bays. This placement strategy ensures that the primary illuminance falls on the driving surface where vehicle movement and pedestrian activity occurs, while minimising shadows cast by parked vehicles. Fittings were distributed with a maximum spacing-to-height ratio of 1.5, giving a maximum spacing of 4.2m at a ceiling height of 2.8m, to achieve acceptable uniformity across the general carpark area. Luminaire positions were additionally offset from structural columns to prevent shadowing on driving surfaces, with the final layout presented in Appendix G.

Illuminance Transition Zones

The carpark entrance lighting was designed as a three-stage transition to facilitate progressive eye adaptation for drivers entering from outdoor daylight conditions. The first 15m zone provides 800 lux, stepping down to 160 lux across the subsequent 4m transition zone, before reducing to the 40 lux general parking level. Each transition maintains an

illuminance ratio within the 10:1 maximum recommended by AS/NZS 1680.1, preventing abrupt visual contrast that could cause temporary blindness or discomfort for drivers.

Day/Night Entrance Lighting Control

AS/NZS 1680 Table D1 clause 11.1 specifies significantly different illuminance requirements for carpark entrances during daytime and nighttime operation. During daytime, the first 15m of the entrance zone requires 800 lux to facilitate driver eye adaptation from high external illuminances. During nighttime, the entrance requirement reduces to 160 lux throughout, with Table D1 clause 11.1(b) explicitly noting that enhanced entry lighting should be automatically reduced at night.

To comply with this requirement, DALI dimmable luminaires (E62/LED4N60D) were specified for both entrance zones alongside standard luminaires (E62/LED2N30S). The dimmable fittings allow the primary lighting level to be automatically adjusted between 800 lux during daytime and 160 lux during nighttime operation, while the standard fittings contribute a consistent base illuminance level throughout the zone. A daylight sensor and timer-based control system manages the dimming automatically, ensuring compliance with both daytime and nighttime requirements without manual intervention.

Generator Room Dual Zone Lighting

The generator room requires two distinct illuminance levels per AS/NZS 1680 Table D1. Clause 9.1 specifies 80 lux for general plant room lighting, while clause 9.2 specifies 160 lux at control panels and switchboards. To address this, general fittings were distributed throughout the room to achieve the 80 lux general requirement, with an additional fitting positioned directly above the switchboard panel to achieve the 160 lux task lighting requirement at that location.

Storage Room Classification

The storage room was classified as live storage, as the space will be actively accessed by staff during normal operations. Per AS/NZS 1680 Table D1 clause 8.1(b), live storage requires a minimum maintained illuminance of 80 lux with a UGR not exceeding 25. The working plane was set at 0.85m to reflect bench-height access to storage shelving, in accordance with AS/NZS 1680.1 recommendations for bench-height tasks.

6.0 Annual CO₂ Emissions Report

6.1 Calculation Methodology

The annual energy consumption of the proposed lighting scheme was obtained from the lighting design calculations, and expressed in kilo-watt hours. The carbon dioxide emissions were then calculated by applying a specified emissions factor of 0.25 kg/kWh to this figure. Two values were calculated, the emissions of the proposed system in buildings A and B, and the potential emissions of the carpark.

6.2 Results for One Year of Operation

The annual reported CO₂ emissions are estimated to be 5,440 kg of CO₂/year, with supporting calculations being found in Appendix D6. This is significantly lower than the existing design, which has annual emissions of 20,978 kg of CO₂/year.

Additionally, the implementation of a new carpark would also generate carbon emissions; the emissions produced by the carpark's lighting system are estimated to be 8,090 kg of CO₂/year. The total carbon dioxide emissions of implementing both the carpark and the proposed lighting systems for building A and B is still lower than the existing design, summing to 13,567 kg of CO₂ /year.

7.0 Design emergency lighting (compliant with F6/F8 clauses of NZBC)

The emergency lighting system has been designed in full compliance with the New Zealand Building Code (NZBC) clauses F6 and F8, and the AS/NZS 2293 series of standards. The system covers all occupied spaces across Buildings A, B, and the carpark, incorporating both escape route lighting and anti-panic lighting. All emergency luminaires are sourced from Legrand and ETAP, selected for their compliance with AS/NZS 2293 and their suitability for the specific spatial requirements of each area. Corridor and anti-panic luminaires operate in non-maintained mode, remaining off during normal operation and activating automatically upon mains power failure, providing a minimum of 90 minutes of illumination in accordance with AS/NZS 2293.1 (Clause 2.2.2). Exit signs operate in maintained mode, remaining illuminated at all times and complying with AS/NZS 2293.3 (Clause 2.4). A maintenance factor of 1.0 has been applied throughout all emergency lighting calculations.

7.1 Reflected Ceiling Plans Buildings A & B

The reflected ceiling plans for Buildings A and B illustrate the placement of all emergency luminaires across the ground floor and first floor of each building. Emergency luminaires are overlaid on the normal lighting reflected ceiling plans in Appendix G. Appendix G2 and Appendix G1 show the emergency luminaire layouts for Building A ground floor and first floor respectively, while Appendix G4 and Appendix G3 show the corresponding layouts for Building B. Escape route lighting is distributed along all designated egress corridors and stairwells, while anti-panic lighting covers open occupied spaces including lobbies, CAD rooms, the kitchen, toilets, and the common room. The luminaire positions have been determined through DIALux Evo simulations to ensure that all illuminance targets are met across every space. The escape routes with pictogram placement for Building A ground floor and first floor are shown in Appendix C2 and Appendix C1 respectively, with Building B ground floor and first floor shown in Appendix C4 and Appendix C3.

7.2 Reflected Ceiling Plan Carpark

The reflected ceiling plan for the carpark presents the emergency lighting layout for the single-level covered parking area. Given the large open-plan nature of the carpark, anti-panic luminaires have been distributed to achieve uniform coverage across all parking bays and internal traffic lanes. Five emergency exit points are identified within the carpark two on the left side and three on the right each marked with a maintained exit sign luminaire. The emergency luminaire overlay for the carpark is shown in Appendix G5, and the corresponding escape route plan with pictogram placement is shown in Appendix C5.

7.3 Compliance with NZBC F6/F8 Requirements

The emergency lighting design satisfies all relevant requirements of NZBC clauses F6 and F8. Under clause F6, all occupied rooms achieve a minimum illuminance of 0.2 lux, and all designated egress paths achieve a minimum of 1.0 lux at floor level, with stairwells achieving a minimum of 2.0 lux at the middle area. The building falls under risk Category C, requiring emergency lighting to remain operational for a minimum of 30 minutes following a mains power failure (F6 Clause 6.3.4); the design exceeds this with a 90-minute battery duration per AS/NZS 2293.1 (Clause 2.2.2). Under clause F8, all emergency exit signs are positioned such that at least one sign is visible from any point along an egress path. The exit sign luminaires used feature lettering of no less than 150 mm in height, ensuring legibility at distances of up to 24 m. Full numerical compliance results for all escape routes and anti-panic zones across Buildings A, B, and the carpark are presented in Appendix E, Table 22.

7.4 Emergency Lighting Design and Pictogram Placement

Emergency exit pictograms have been placed at all five exit points throughout the carpark and at all final exits for both floors in Buildings A and B in accordance with NZBC F8 and AS/NZS 2293.3. The pictograms conform to ISO 7010 E001/E002, featuring the standard green background with a white running figure and a directional arrow indicating the nearest egress route. Single-sided signs are used where exits are visible directly ahead or to one side, with directional arrows oriented to guide occupants toward the closest exit. For the carpark, exits on the left wall display arrows pointing left, while exits on the right wall display arrows pointing right, ensuring intuitive wayfinding under emergency conditions. The full escape route plans indicating pictogram placement for Buildings A and B are shown in Appendix C1 through C4, and the carpark escape route plan is shown in Appendix C5. Compliance of all escape routes and anti-panic zones against the illuminance and uniformity requirements of AS/NZS 2293 is confirmed in Appendix E, Table 22.

8.0 Conclusion

In conclusion, the lighting design proposal for Alpha Design Limited's two office buildings and new covered carpark has delivered a comprehensive evaluation of the existing system and a fully redesigned solution that meets all relevant New Zealand standards and client requirements. The existing fluorescent lighting across Buildings A and B was found to be non-compliant with AS/NZS 1680 in the majority of rooms, with uniformity failures, excessive glare, and a power density of 21.68 W/m² significantly exceeding the recommended 7-9 W/m² for office environments. The new LED-based design addresses these deficiencies across all spaces, achieving compliant illuminance, uniformity, and UGR values throughout both buildings and the carpark.

The carpark lighting has been designed from scratch as a complete new installation, meeting the tiered illuminance requirements of AS/NZS 1680 Table D1 across all zones, including entrance, transition, general parking, storage, and generator areas. DALI dimming controls have been incorporated at both entrance zones to automatically manage the daytime and nighttime illuminance requirements without manual intervention.

The emergency lighting system has been designed in full compliance with NZBC clauses F6 and F8 and AS/NZS 2293, providing a minimum of 90 minutes of illumination across all escape routes and anti-panic zones in Buildings A, B, and the carpark. Maintained exit signs conforming to ISO 7010 are positioned at all five carpark exit points and all final exits across both buildings, ensuring clear and compliant wayfinding under emergency conditions.

The proposed LED system achieves a payback period of approximately 6 years, well within the 10-year maximum set by the client brief, with a significantly reduced power density and annual CO₂ emissions compared to the existing installation. The recycling cost of \$6,780 for the existing fluorescent lamps further reinforces the long-term financial and environmental case for transitioning to LED technology. Overall, the redesigned lighting system delivers a safe, efficient, and compliant solution that supports Alpha Design Limited's operational needs while positioning the business for long-term energy and cost savings.

This project also aligns with several United Nations Sustainable Development Goals. The transition to LED technology directly supports SDG 7 (Affordable and Clean Energy) by reducing total energy consumption by 74%, and SDG 13 (Climate Action) through a reduction in annual CO₂ emissions from 20.98 tonnes to 5.48 tonnes per year. The responsible disposal and recycling of the existing 678 fluorescent lamps, along with the selection of longer-life LED luminaires that reduce future waste, reflects the principles of SDG 12 (Responsible Consumption and Production).

Finally, the team considered the principles of Te Tiriti o Waitangi throughout this project. The principle of Partnership was reflected in our collaborative approach with Legrand New Zealand as an industry partner and with Alpha Design Limited as the client, ensuring all stakeholders were engaged in the design process. Participation was upheld by designing inclusive, safe, and well-lit spaces accessible to all building users, including provisions for disabled toilet facilities and clearly marked emergency egress routes. Protection was considered through the significant reduction in CO₂ emissions and responsible lamp recycling, acknowledging our obligation to protect the environment taonga for future generations.

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10.0 Appendices

Appendix A: Evaluation of Existing Lighting Results

Building A - Ground Floor

101 Stairs Calculation Surface	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
Floor	206		0.84		
Staircase slope	239		0.14		
Top step	1.63		0.6		
Average	149	≥ 80	0.53	≥ 0.3	Yes

Table 1: Lighting Compliance Table of Building A 101 Stairs

Rooms	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
102 Corridor/Lobby	501	≥ 80	0.46	≥ 0.3	Yes
103 Toilet	207	≥ 80	0.76	≥ 0.5	Yes
104 Toilet	208	≥ 80	0.78	≥ 0.5	Yes
105 Toilet - Disabled	187	≥ 80	0.75	≥ 0.5	Yes
106 Meeting	625	≥ 320	0.58	≥ 0.5	Yes
107 Meeting	644	≥ 320	0.59	≥ 0.5	Yes
108 Corridor	373	≥ 40	0.22	≥ 0.3	No
109 Mail	614	≥ 80	0.45	≥ 0.5	No
110 Office	836	≥ 320	0.51	≥ 0.5	Yes
111 CAD	743	≥ 320	0.54	≥ 0.5	Yes
112 CAD	1088	≥ 320	0.5	≥ 0.5	Yes
113 Office	744	≥ 320	0.41	≥ 0.5	No
114 Visitor	954	≥ 160	0.5	≥ 0.5	Yes

Table 2: Lighting Compliance Table of Building A Ground Floor

Room	UGR	UGR - Standard	Meets Compliance
111 CAD:	19.4	≤ 19	No
112 CAD	19.5	≤ 19	No

Table 3: Glare Compliance Table of Building A Ground Floor

Building A - First Floor

Room	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
201 Stairs	140	≥ 80	0.76	≥ 0.3	Yes
202 Corridor	141	≥ 40	0.49	≥ 0.3	Yes
203 Meeting	330	≥ 320	0.85	≥ 0.5	Yes
204 Manager	469	≥ 320	0.54	≥ 0.5	Yes
204A DB	0	≥ 160	0	≥ 0.5	No
205 Balcony	144	≥ 40	0.34	≥ 0.3	Yes
206 Toilet	266	≥ 80	0.79	≥ 0.5	Yes
207 Office	517	≥ 320	0.58	≥ 0.5	Yes
208 Meeting	459	≥ 320	0.79	≥ 0.5	Yes
209 Meeting	424	≥ 320	0.81	≥ 0.5	Yes
210 CAD	455	≥ 320	0.67	≥ 0.5	Yes
211 Corridor	132	≥ 40	0.41	≥ 0.3	Yes
212 Common	423	≥ 160	0.66	≥ 0.5	Yes
213 Balcony	98	≥ 40	0.24	≥ 0.5	No
214 Office	413	≥ 320	0.73	≥ 0.5	Yes
215 Meeting	419	≥ 320	0.74	≥ 0.5	Yes
216 Meeting	400	≥ 320	0.68	≥ 0.5	Yes
217 Toilet	139	≥ 80	0.78	≥ 0.5	Yes
218 Toilet	201	≥ 80	0.73	≥ 0.5	Yes
219 Meeting	399	≥ 320	0.71	≥ 0.5	Yes
220 Office	389	≥ 320	0.76	≥ 0.5	Yes

Table 4: Lighting Compliance Table of Building A First Floor

Room	UGR	UGR - Standard	Meets Compliance
210 CAD	22	≤ 19	No

Table 5: Glare Compliance Table of Building A First Floor

Building B - Ground Floor

101 Stairs Calculation Surface	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
Floor	207		0.86		
Staircase slope	110		0.10		
Top step	2.04		0.38		
Average	106	≥ 80	0.45	≥ 0.3	Yes

Table 6: Lighting Compliance Table of Building B 101 Stairs

Room	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
102 Corridor	380	≥ 40	0.25	≥ 0.3	No
103 Storage	321	≥ 40	0.38	≥ 0.3	Yes
104 Utility	199	≥ 40	0.77	≥ 0.3	Yes
105 WC	200	≥ 80	0.74	≥ 0.5	Yes
106 WC	185	≥ 80	0.72	≥ 0.5	Yes
107 Shower	185	≥ 80	0.73	≥ 0.5	Yes
108 Office	753	≥ 320	0.43	≥ 0.5	No
109 Office	789	≥ 320	0.47	≥ 0.5	No
110 CAD	1107	≥ 320	0.34	≥ 0.5	No
111 Manager	740	≥ 320	0.5	≥ 0.5	Yes
112 Meeting	865	≥ 320	0.35	≥ 0.5	No
113 Corridor/Coffee	401	≥ 160	0.55	≥ 0.5	Yes
114 Storage	404	≥ 40	0.68	≥ 0.3	Yes
115 Reception	1069	≥ 320	0.52	≥ 0.5	Yes

116 Meeting	854	≥ 320	0.59	≥ 0.5	Yes
117 Corridor	538	≥ 40	0.41	≥ 0.3	Yes
118 Office	1064	≥ 320	0.57	≥ 0.5	Yes
119 Meeting	538	≥ 320	0.49	≥ 0.5	No
120 Office	1133	≥ 320	0.56	≥ 0.5	Yes
121 Meeting	900	≥ 320	0.54	≥ 0.5	Yes

Table 7: Lighting Compliance Table of Building B Ground Floor

Room	UGR	UGR - Standard	Meets Compliance
110 CAD	18.1	≤ 19	Yes

Table 8: Glare Compliance Table of Building B Ground Floor

Building B - First Floor

Room	Eave (lux)	Eave (lux) - Standard	Uo	Uo - Standard	Meets Compliance
201 Stairs	237	≥ 80	0.84	≥ 0.3	Yes
202 Corridor	169	≥ 40	0.049	≥ 0.3	No
203 Director	291	≥ 320	0.66	≥ 0.5	Yes
203a DB	0	≥ 160	0	≥ 0.5	No
204 Meeting	387	≥ 320	0.75	≥ 0.5	Yes
205 Office	375	≥ 320	0.6	≥ 0.5	Yes
206 Kitchen	304	≥ 240	0.57	≥ 0.5	Yes
207 CAD	476	≥ 320	0.66	≥ 0.5	Yes
208 Meeting	554	≥ 320	0.73	≥ 0.5	Yes
209 Toilet	223	≥ 80	0.74	≥ 0.5	Yes
210 Office	678	≥ 320	0.69	≥ 0.5	Yes
211 Toilet	270	≥ 80	0.81	≥ 0.5	Yes
212 Office	622	≥ 320	0.7	≥ 0.5	Yes
212a DB	138	≥ 160	0.72	≥ 0.5	No
213 Conference	456	≥ 320	0.59	≥ 0.5	Yes

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Table 9: Lighting Compliance Table of Building B First Floor

Room	UGR	UGR - Standard	Meets Compliance
207 CAD	21.4	<= 19	No

Table 10: Glare Compliance Table of Building B First Floor

Appendix B: New Design Results

B1. Building A First Floor

Room	Eav (lx)	Eav (lx) - Standard	Uniformity	Uniformity-Standard	Meets Compliance
201 Stairs	90.5	80	0.75	0.3	Yes
202 Corridor	87.7	40	0.46	0.3	Yes
203 Meeting	348	320	0.85	0.5	Yes
204 Manager	462	320	0.85	0.5	Yes
205 Balcony	144	40	0.47	0.3	Yes
206 Toilet	120	80	0.92	0.5	Yes
207 Office	346	320	0.61	0.5	Yes
208 Meeting	368	320	0.82	0.5	Yes
209 Meeting	345	320	0.81	0.5	Yes
210 CAD	444	320	0.68	0.5	Yes
211 Corridor	140	40	0.4	0.3	Yes
212 Common	179	160	0.67	0.5	Yes
213 Balcony	154	40	0.52	0.3	Yes
214 Office	325	320	0.73	0.5	Yes
215 Meeting	337	320	0.73	0.5	Yes
216 Meeting	333	320	0.71	0.5	Yes
217 Toilet	86.4	80	0.91	0.5	Yes
218 Toilet	88.9	80	0.72	0.5	Yes
219 Meeting	374	320	0.8	0.5	Yes
220 Office	353	320	0.77	0.5	Yes

Table 11: Lighting Compliance Table of Building A First Floor (NEW)

Room	UGR	UGR - Standard	Meets Compliance
203 Meeting	13.1	≤ 19	Yes
204 Manager	16.3	≤ 19	Yes
207 Office	15.9	≤ 19	Yes
208 Meeting	16.3	≤ 19	Yes
209 Meeting	16.7	≤ 19	Yes
210 CAD	17.2	≤ 19	Yes
214 Office	15.2	≤ 19	Yes
215 Meeting	16	≤ 19	Yes
216 Meeting	17.4	≤ 19	Yes
219 Meeting	13.8	≤ 19	Yes
220 Office	15.9	≤ 19	Yes

Table 12: Glare Compliance Table of Building A First Floor (NEW)

B2. Building B First Floor

Room	Eav (lx)	Eav (lx) - Standard	Uniformity	Uniformity-Standard	Meets Compliance
201 Stairs	104	80	0.52	0.3	Yes
202 Corridor	117	40	0.39	0.3	Yes
203 Director	385	320	0.75	0.5	Yes
203 DB	229	160	0.91	0.5	Yes
204 Meeting	372	320	0.77	0.5	Yes
205 Office	416	320	0.70	0.5	Yes
206 Kitchen	379	240	0.73	0.5	Yes
207 CAD	539	320	0.75	0.5	Yes
208 Meeting	509	320	0.74	0.5	Yes
209 Toilet	173	80	0.61	0.5	Yes
210 Office	343	320	0.63	0.5	Yes
211 Toilet	252	80	0.62	0.5	Yes

212 Office	488	320	0.75	0.5	Yes
212A DB	243	160	0.69	0.5	Yes
213 Conference	556	240	0.72	0.5	Yes

Table 13: Lighting Compliance Table of Building B First Floor (NEW)

Room	UGR	UGR - Standard	Meets Compliance
203 Director	17.1	<=19	Yes
204 Meeting	16.8	<=19	Yes
205 Office	18.7	<=19	Yes
206 Kitchen	17.2	<=19	Yes
207 CAD	16.4	<= 19	Yes
208 Meeting	18.2	<=19	Yes
210 Office	18.2	<=19	Yes
212 Office	16.3	<=19	Yes
213 Conference	18.7	<=19	Yes

Table 14: Glare Compliance Table of Building B First Floor (NEW)

B3. Building A Ground Floor

Room	Eav (lx)	Eav (lx) - Standard	Uniformity	Uniformity-Standard	Meets Compliance
101 Stairs	143.95	80	0.37	0.3	Yes
102 Corridor/Lobby	137	40	0.47	0.3	Yes
103 Toilet	164	80	0.76	0.5	Yes
104 Toilet	147	80	0.78	0.5	Yes
105 Toilet Disable	127	80	0.7	0.5	Yes
106 Meeting	353	320	0.58	0.5	Yes
107 Meeting	381	320	0.59	0.5	Yes
108 Corridor	164	40	0.3	0.3	Yes
109 Mail	355	320	0.5	0.5	Yes

110 Office	348	320	0.68	0.5	Yes
111 CAD	643	320	0.51	0.5	Yes
112 CAD	692	320	0.53	0.5	Yes
113 Office	488	320	0.69	0.5	Yes
114 Visitors	166	160	0.7	0.5	Yes

Table 15: Lighting Compliance Table of Building A Ground Floor (NEW)

Room	UGR	UGR - Standard	Meets Compliance
106 Meeting Room	16.7	≤ 19	Yes
107 Meeting Room	14.6	≤ 19	Yes
109 Office / Mail Room	16.3	≤ 19	Yes
110 Office	16.2	≤ 19	Yes
111 CAD / Drafting	16.0	≤ 19	Yes
112 CAD / Drafting	18.1	≤ 19	Yes
113 Office	15.0	≤ 19	Yes

Table 16: Glare Compliance Table of Building A Ground Floor (NEW)

B4. Building B Ground Floor

Room	Eav (lx)	Eav (lx) - Standard	Uniformity	Uniformity-Standard	Meets Compliance
101 Stairs	290	80	0.71	0.3	Yes
102 Corridor	179	40	0.30	0.3	Yes
103 Storage	437	40	0.54	0.5	Yes
104 Utility	318	40	0.69	0.5	Yes
105 WC	354	80	0.60	0.5	Yes
106 WC	306	80	0.56	0.5	Yes
107 Shower	210	80	0.58	0.5	Yes

108 Office	659	320	0.52	0.5	Yes
109 Office	351	320	0.52	0.5	Yes
110 CAD	765	320	0.52	0.5	Yes
111 Manager	330	320	0.59	0.5	Yes
112 Meeting	456	320	0.63	0.5	Yes
113 Corridor/Coffee	401	40	0.55	0.3	Yes
114 Storage	253	40	0.59	0.5	Yes
115 Reception	462	240	0.55	0.5	Yes
116 Meeting	365	320	0.50	0.5	Yes
117 Corridor	320	40	0.38	0.3	Yes
118 Office	442	320	0.51	0.5	Yes
119 Meeting	370	320	0.59	0.5	Yes
120 Office	492	320	0.52	0.5	Yes
121 Meeting	346	320	0.63	0.5	Yes

Table 17: Lighting Compliance Table of Building B Ground Floor (NEW)

Room	UGR	UGR - Standard	Meets Compliance
108 Office	18.5	≤ 19	Yes
109 Office	18.4	≤ 19	Yes
110 CAD	18.1	≤ 19	Yes
111 Manager	18.8	≤ 19	Yes
112 Meeting	17.3	≤ 19	Yes
116 Meeting	17.9	≤ 19	Yes
118 Office	17.8	≤ 19	Yes
119 Meeting	18.4	≤ 19	Yes
120 Office	18.8	≤ 19	Yes
121 Meeting	16.0	≤ 19	Yes

Table 18: Glare Compliance Table of Building B Ground Floor (NEW)

B5. Carpark

Room	Eav (lx)	Eav (lx) - Standard	Uniformity	Uniformity-Standard	Meets Compliance
First 15 meters	816	800	0.54	0.25	Yes
Next 4 meters	190	160	0.38	0.25	Yes
Rest of Carpak	110	40	0.29	0.25	Yes
Generator room	114	80	0.51	0.5	Yes
Storage room	118	80	0.69	0.5	Yes

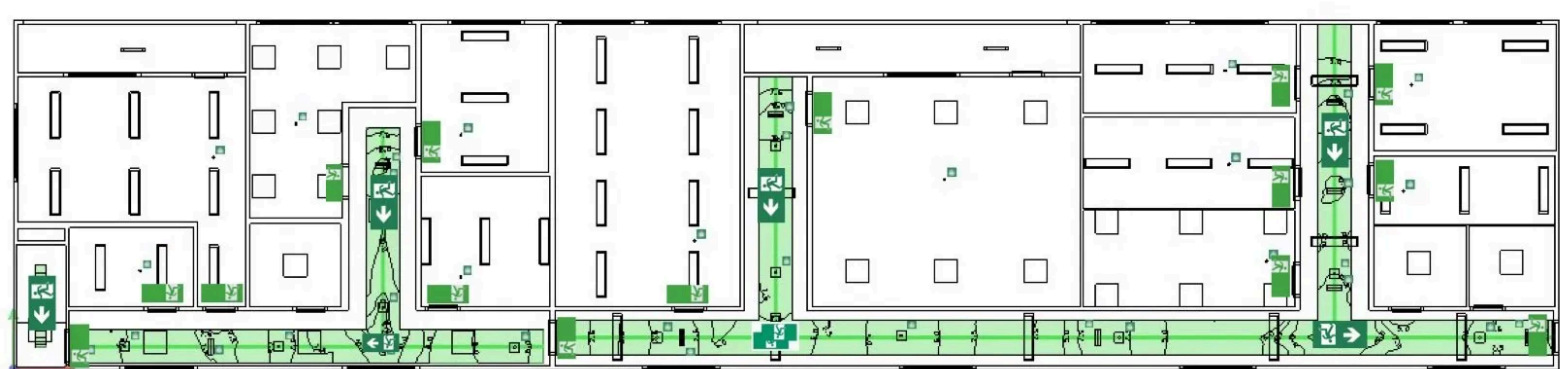
Table 19: Lighting Compliance Table of Carpark

Room	UGR	UGR - Standard	Meets Compliance
Generator room	23	<= 25	Yes
Storage room	24	<= 25	Yes

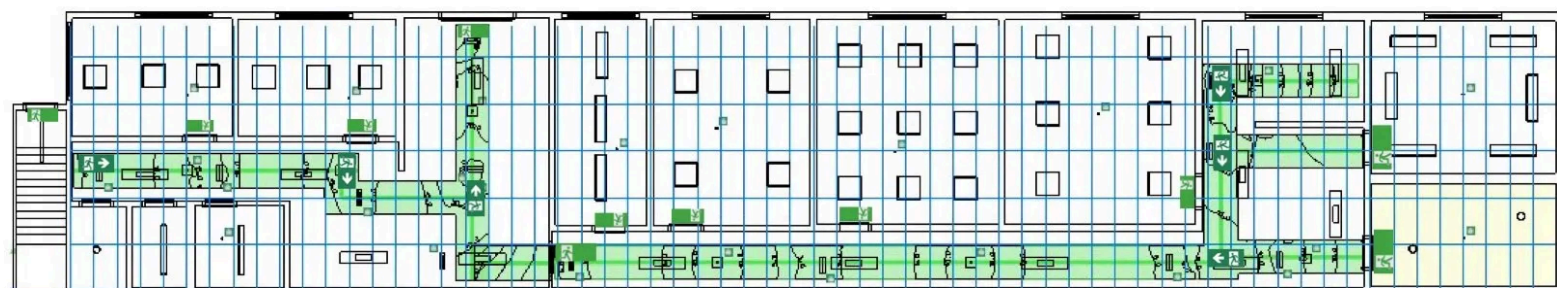
Table 20: Lighting Compliance Table of Carpark

Appendix C: Escape Routes with Pictograms

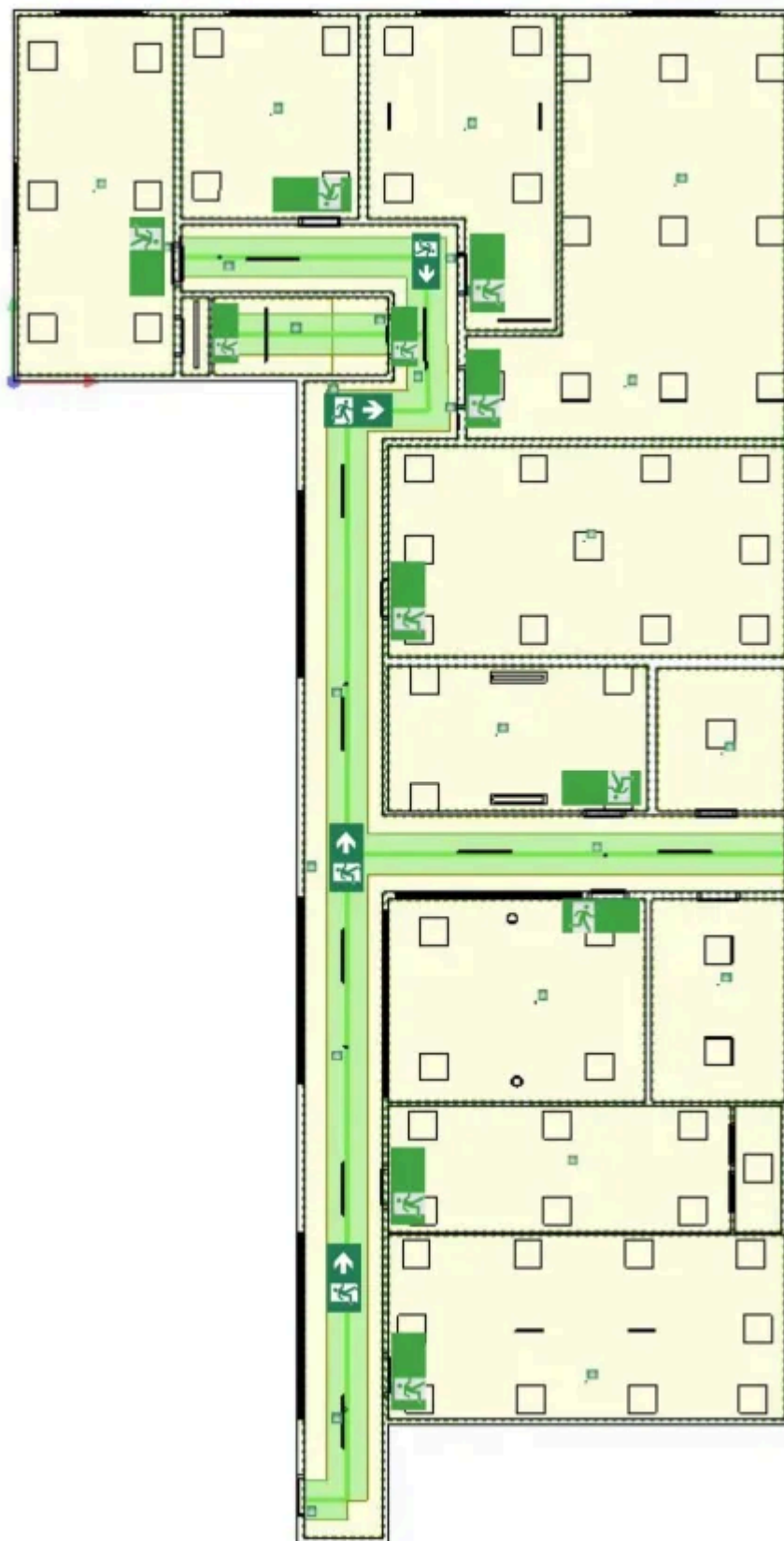
C1. Building A first floor



C2. Building A ground floor



C3. Building B first Floor



C4. Building B ground floor



C5. Carpark



Appendix D: Calculations

D1: Recycling Cost Calculations

Luminaire list

Φ_{total} 1491885 lm	P_{total} 25428.0 W	Luminous efficacy 58.7 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
171	Philips	2x36W		76.0 W	4663 lm	61.4 lm/W
84	Philips	4x36W SLB		148.0 W	8268 lm	55.9 lm/W

Figure 1B: Existing lighting system Luminaire list

T8 Lamp count = $171 \times 2 + 84 \times 4 = 678$

Recycling cost = $\$10 \times 678 = \mathbf{\$6780}$

D2: Cost Summary

Office hours: 12 hours a day, 5.5 day per week, 50 weeks per year = 3300 hours/year

For buildings A and B:

For existing design:

Annual energy consumption: $(171 \times 76 + 84 \times 148) \times 3300 = 83912.4$ kWh/year

Annual energy cost of existing system: $83912.4 \text{ kWh/year} * \$0.25 = \text{\$20978.1 /year}$

For proposed design:

Annual energy consumption: $(1723+879+2143+1894)*3300 = 21908.7 \text{ kWh/year}$

Annual energy cost of proposed system: $21908.7 \text{ kWh/year} * \$0.25 = \text{\$5477.2 /year}$

Annual savings for buildings A&B: $\text{\$20978} - \text{\$5477} = \text{\$15501}$

D3: Power Density of New Design

For existing design:

Total area of Building A + Building B: 1173 m^2

Found by finding the area of each room in Dialux then summed together.

Total building power: 25428.0 W

Power density (EXISTING) = $25428.0 \text{ W} / 1173 \text{ m}^2 = \textbf{21.68 W/m}^2$

For new design:

Total building power: $(1723+879+2143+1894) = 6639 \text{ W}$

Power density (NEW) = $6639 \text{ W} / 1173 \text{ m}^2 = \textbf{5.66 W/m}^2$

D4: Payback Time Calculations

Payback time has been calculated by fitting design parameters to a simple linear equation to find the annual costs of the 4 individual floors. Given that the cost of each new luminaire is \$300 and the net energy price is \$0.25/kWh, and office lighting hours: 12 hours a day, 5.5 day per week, 50 weeks per year = 3300 hours, the equation below was used.

For the equation: $C = 0.25*3300*10^{-3}*Wx + 300n$

Let W = the total watts used by all luminaires per year on the given floor

x = the number of years passed

n = the total number of new Luminaires on a given floor

C = the running cost of the lighting system

For the existing system $n = 0$ in all cases, so the cost is only dependent on the total watts used and the number of years, whereas for the proposed system $n > 0$, so there is a fixed offset that represents the initial purchase price.

The parameter values for each floor in the proposed system are as follows:

- Building A First Floor:
 - $n = 74$
 - $W = 1723$
- Building A Ground Floor:
 - $n = 53$
 - $W = 879$
- Building B First Floor:
 - $n = 80$

- W = 2097.9
- Building B Ground Floor:
- n = 71
- W = 1894

Summing these up we find that

Total cost to purchase new luminaires = **\$83,400**

Total cost per year to run proposed system = **\$5,440**

The total cost of the existing and proposed systems for the first ten years is shown in the table below. Payback is achieved in year ~6.

TOTAL		
Year #	Current Lighting System C (\$)	New Lighting System C (\$)
1	20978	88840
2	40274	94280
3	60411	99720
4	80548	105160
5	100685	110600
6	120822	116040
7	140959	121480
8	161096	126920
9	181233	132360
10	201370	137800

Table 21: Total Running Cost of Existing vs Proposed systems

D5: Maintenance Factor Calculations

$$MF = LLMF \times LMF \times LSF \times RSMF$$

Where MF is the Maintenance Factor, LLMF is the Lamp Lumen Maintenance Factor, LMF is the Luminaire Maintenance Factor LSF is the Lamp Survival Factor RSMF is the Room Surface Maintenance Factor LLMF is obtained through the datasheet LMF is reasonably assumed at 0.95 to account for dirt/dust buildup LSF is 1, because it is assumed that once a luminaire breaks it is replaced RSMF will be determined using Room Index, where Room Index is calculated as:

$$LMF = 0.95$$

$$LSF = 1$$

$$RI = \frac{L \times W}{L + W} (H - h)$$

Where RI is the Room Index, L is the Length of the Room, W is the Width of the Room, H is the Height of the Room, h is the Height of the Working plane. The room index was then compared to the RSMF Table, to find the corresponding RSMF.

RSMF = around 0.94 taken from this table:

Room Index	Cleaning Every Year (Clean)	Normal	Dirty	Cleaning Every 2 Years (Clean)	Normal	Dirty	Cleaning Every 3 Years (Clean)	Normal	Dirty
Small Room (RI = 0.7)	0.97	0.94	0.93	0.95	0.93	0.90	0.94	0.92	0.88
Medium Room (RI = 2.5)	0.98	0.96	0.95	0.96	0.95	0.94	0.96	0.95	0.94
Large Room (RI = 5)	0.98	0.96	0.95	0.96	0.95	0.94	0.96	0.95	0.94

Table 22: RSMF Table

D6: CO₂ Emissions Calculations

Office hours: 12 hours a day, 5.5 day per week, 50 weeks per year = 3300 hours/year

Carpark lighting: 24 hours a day, 7 days per week, 52 weeks per year = 8736 hours/year

For existing design:

Total power of building: 25428 W = 25.428 kW

Annual energy consumption: 25.428 kW x 3300h = 83912.4 kWh/year

CO₂ emission: 0.25kg/kWh

Annual CO₂ emission = Annual energy consumption x CO₂ emission = 83912.4 kWh/year x 0.25kg/kWh = **20978.1 kg CO₂ / year** = 20.98 tonnes of CO₂ / year

For proposed design:

Total power of buildings A and B: 6593.9 W = 6.5939 kW

Annual energy consumption: 6.5939 kW x 3300h = 21759.9 kWh/year

CO2 emission: 0.25kg/kWh

Annual CO2 emission = 21759.9 kWh/year x 0.25kg/kWh = **5440.0 kg CO2 / year**

Carpark:

Refer to Appendix F2 for used values of wattage.

Annual energy consumption: $(3 \times 26.5 + 50 \times 25 + 50 \times 47.5) \times 10^{-3} \text{ kW} \times 8736 = 32362.5 \text{ kWh/year}$

Annual CO2 emission = 32362.5 kWh/year x 0.25kg/kWh = **8090 kg CO2 / year**

Appendix E: Emergency Lighting Results

For escape routes to comply with standards:

Emin center line for anti panic surfaces > 0.2 lx

Emin middle area of stairs > 2 lx

Emin middle area excluding stairs > 1 lx

Emin centre line excluding stairs > 1 lx

Uniformity > 0.025

Room	Emin (lx) centre line	Emin (lx) middle area	uniformity	Meets Compliance
Building A - Ground Floor				
101 Stairs	2.51	2.13	0.67	Yes
102 Corridor/coffee	2.22	1.95	0.74	Yes
108 Corridor	2.38	2.09	0.45	Yes
Building A - First Floor				
201 Stairs	4.35	3.92	0.57	Yes
202 Corridor	2.35	1.88	0.34	Yes
211 Corridor	2.65	2.13	0.30	Yes
Building B - Ground				
101 Stairs - Floor	5.65	5.19	0.85	Yes
101 Stairs - Staircase	3.06	2.78	0.31	Yes

102 Corridor	2.83	2.18	0.18	Yes
113 Corridor/Coffee	2.64	1.68	0.41	Yes
117 Corridor	2.44	2.43	0.15	Yes
Building B - First Floor				
201 Stairs - Top Floor	3.49	2.81	0.65	Yes
202 Corridor	1.79	1.67	0.15	Yes
Carpark				
Shared open area (escape route)	3.8	3.1	0.29	Yes

Table 23: Emergency route Compliance Table

Appendix F: Luminaire Schedule

F1: Buildings A and B Luminaire List

Building A - FIRST FLOOR

Luminaire list

Φ_{total}	P_{total}	Luminous efficacy	$\Phi_{Emergency\ lighting}$	$P_{Emergency\ lighting}$
212764 lm	1835.6 W	115.9 lm/W	7514 lm	113.1 W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
3	ETAP	-	E2A-S102-1P-840-030D (_LTrvAA)	18.5 W	2695 lm	145.6 lm/W
13	Legrand	659009LI	LED SATELLITE D63 WHITE GALAXY	3.8 W	305 lm	80.2 lm/W
				 3.8 W	305 lm (100 %)	-
13	Legrand	659017LI	LED SATELLITE CORRIDOR SURFACE WHITE GALAXY	2.5 W	215 lm	85.9 lm/W
				 2.5 W	215 lm (100 %)	-
13	Legrand	659118LI	EDGE LIGHT REC SS RML LI GAL	2.4 W	58 lm	24.5 lm/W
				 2.4 W	58 lm (100 %)	-
27	Lena Lighting	021046	ARTO 2 LED 2250LM I KL. IP20 592X592X45MM 840 21W	21.0 W	2250 lm	107.1 lm/W
44	Lena Lighting	902635	COMPACT LED EVO N 3100LM NPRM UGR II KL. IP20 1200X250 840 (25W)	25.0 W	3100 lm	124.0 lm/W

Figure N1: Proposed system Building A First Floor Luminaire list

Building B First Floor:

Normal lighting

1	Lena Lighting	514890	TABLO LED P AS 4650lm 840 (35W)	38.4 W	4650 lm	121.1 lm/W
4	Lena Lighting	630644	COMPACT LED EVO P ZK 3550lm PRM 840 (24W)	25.0 W	3550 lm	142.0 lm/W
10	Lena Lighting	630651	COMPACT LED EVO P ZK 3300lm PRM 830 (24W)	25.0 W	3300 lm	132.0 lm/W
49	Lena Lighting	239717	TERRA 2 LED ZW 595X595MM X4 840 BIALY MAT (MULTIWATT 24-36W)	24.0 W	3000 lm	125.0 lm/W
2	Lena Lighting	769962	BARIS 40 LED UGR PLUS N 2500lm I kl. IP20 1140mm 840 19W BIAEY 70D	19.0 W	2500 lm	131.6 lm/W
2	Lena Lighting	238192	TERRA 2 LED LONG Z 2x1 1195x230mm 3000lm 840 CZARNY MAT (36W)	39.8 W	3000 lm	75.4 lm/W
2	Lena Lighting	319228	BARIS 40 LED N VD 1100lm PLX I KL. IP20 579mm 840 ANODA CO 15W	15.0 W	1100 lm	73.3 lm/W
pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
10	Lena Lighting	319426	BARIS 40 LED N VD 2500lm PLX I KL. IP20 1140mm 840 BIALY 32W	32.0 W	2500 lm	78.1 lm/W

Emergency lighting

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
11	ETAP	-	K132/3 (_LTRevAB)	1.5 W	45 lm	30.0 lm/W
				 1.5 W	45 lm (100 %)	-
6	Legrand	659017LI	LED SATELLITE CORRIDOR SURFACE WHITE GALAXY	2.5 W	215 lm	85.9 lm/W
				 2.5 W	215 lm (100 %)	-
13	Legrand		659326GA 659326SA 659327GA 659327SA 659328GA 659328SA 659329GA 659329SA	1.7 W	268 lm	159.6 lm/W
				 1.7 W	268 lm (100 %)	-

Building A Ground Floor Normal and Emergency Lighting (green symbol):

Building A · GROUND FLOOR

Luminaire list

Φ_{total} 145852 lm		P_{total} 1111.0 W	Luminous efficacy 131.3 lm/W		$\Phi_{\text{Emergency lighting}}$ 5376 lm	$P_{\text{Emergency lighting}}$ 85.5 W	
pcs.	Manufacturer	Article No.	Article name		P	Φ	Luminous efficacy
2	ETAP	-	D1C-R300-11-840-015D (_LTRRevAA)		10.0 W	1447 lm	144.7 lm/W
1	ETAP	-	D9B-R310-10-840-015D (_LTRRevAA)		10.0 W	1450 lm	145.0 lm/W
1	ETAP	-	R820R1/LEDN25D (_LTRRevAC)		18.0 W	2499 lm	138.8 lm/W
9	ETAP	-	U2B-M110-MI-840-025D (_LTRRevAD)		17.0 W	2495 lm	146.7 lm/W
19	ETAP	-	U2B-M210-MI-840-025D (_LTRRevAD)		16.5 W	2495 lm	151.2 lm/W
6	ETAP	-	U2B-M210-MI-840-045S (_LTRRevAD)		29.5 W	4490 lm	152.2 lm/W
13	ETAP	-	U7C-M112-35-840-020D (_LTRRevAG)		14.0 W	2097 lm	149.8 lm/W
9	Legrand	659009LI	LED SATELLITE D63 WHITE GALAXY		3.8 W	305 lm	80.2 lm/W
					3.8 W	305 lm (100 %)	-
9	Legrand	659017LI	LED SATELLITE CORRIDOR SURFACE WHITE GALAXY		2.5 W	215 lm	85.9 lm/W
					2.5 W	215 lm (100 %)	-
12	Legrand	659118LI	EDGELIGHT REC SS RML LI GAL		2.4 W	58 lm	24.5 lm/W
					2.4 W	58 lm (100 %)	-

Building B Ground Floor:

Building B · GROUND FOOR

Luminaire list

Φ_{total} 225840 lm		P_{total} 1899.0 W		Luminous efficacy 118.9 lm/W		
pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
9	Lena Lighting	021008	ARTO 2 LED 3950LM I KL. IP20 592X592X45MM 840 37W	37.0 W	3950 lm	106.8 lm/W
10	Lena Lighting	021046	ARTO 2 LED 2250LM I KL. IP20 592X592X45MM 840 21W	21.0 W	2250 lm	107.1 lm/W
2	Lena Lighting	021053	ARTO 2 LED 5270LM I KL. IP20 592X592X45MM 840 53W	53.0 W	5270 lm	99.4 lm/W
5	Lena Lighting	630644	COMPACT LED EVO P ZK 3550lm PRM 840 (24W)	25.0 W	3550 lm	142.0 lm/W
45	Lena Lighting	902635	COMPACT LED EVO N 3100LM NPRM UGR II KL. IP20 1200X250 840 (25W)	25.0 W	3100 lm	124.0 lm/W

F2: Carpark Luminaire List

Carpark Normal and Emergency Lighting (green symbol)

Carpark

Luminaire list

Φ_{total} 479005 lm		P_{total} 3913.4 W		Luminous efficacy 122.4 lm/W		$\Phi_{Emergency\ lighting}$ 17355 lm	$P_{Emergency\ lighting}$ 208.9 W
pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy	
3	ETAP		E62/LED2N30D	26.5 W	3049 lm	115.0 lm/W	
50	ETAP		E62/LED2N30S	25.0 W	3049 lm	121.9 lm/W	
50	ETAP		E62/LED4N60D	47.5 W	5997 lm	126.3 lm/W	
41	Legrand	659017LI	LED SATELLITE CORRIDOR SURFACE WHITE GALAXY	2.5 W	215 lm	85.9 lm/W	
				 2.5 W	215 lm (100 %)	-	
28	Legrand	659504LI	LED SATELLITE D63 SURFACE WHITE	3.8 W	305 lm	80.2 lm/W	
				 3.8 W	305 lm (100 %)	-	

F3: Emergency Lighting Luminaire Schedule

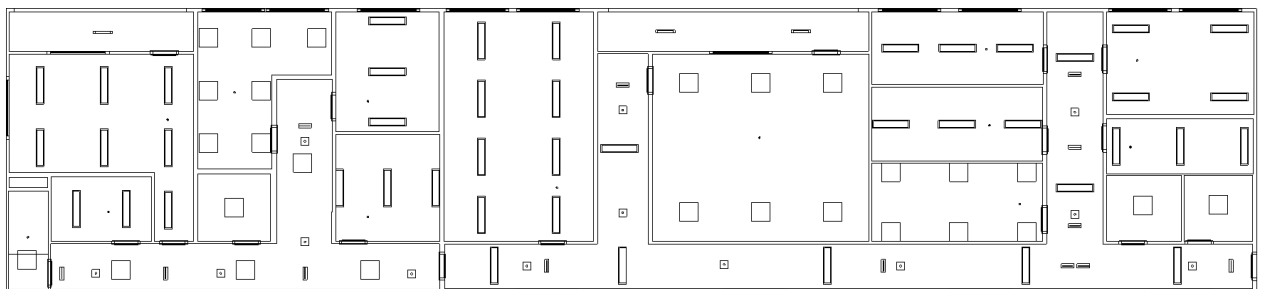
Key	Photo	Product	Dimension	Mounting Type	Flux/Colour	Power	LLMF @ 50 khrs
a		Legrand LED SATELLITE CORRIDOR SURFACE WHITE GALAXY	251mm x 251mm x 60mm	Surface Mount	215 lm / 3000K	2.5W	1
b		Legrand D50 LED Surface Mounted Galaxy Integrated Satellite White	251mm x 251mm x 60mm	Surface Mount	284 lm / 5000k	3.8W	1
c		Legrand LED SATELLITE D63 SURFACE WHITE	251mm x 251mm x 60mm	Surface Mount	305 lm / 3000K	2W	1
d		659044LI Securit Galaxy LED Recessed 4W Emergency Exit Sign - Single Sided Running Man STRAIGHT	406mm x 112mm x 267mm	Recessed	16/ 3000k	4W	1
e		659045LI Securit Galaxy LED Recessed 4W Emergency Exit Sign - Single Sided Running Man LEFT	406mm x 112mm x 267mm	Recessed	16/ 3000k	4W	1
f		659046LI Securit Galaxy LED Recessed 4W Emergency Exit Sign - Single Sided Running Man RIGHT	406mm x 112mm x 267mm	Recessed	16/ 3000k	4W	1

g		659047LI Securit Galaxy LED Recessed 4W Emergency Exit Sign - Double Sided Running Man STRAIGHT	406mm x 112mm x 267mm	Recessed	16/ 3000k	4W	1
h		659048LI Securit Galaxy LED Recessed 4W Emergency Exit Sign - Double Sided Running Man LEFT / RIGHT	406mm x 112mm x 267mm	Recessed	16/ 3000k	4W	1
		Legrand Edgelight K132/3 - K1EL1	304 mm x 155 mm x 73 mm	Wall Mounted	45 lm / 6000k	1.5W	1
		Legrand Edgelight K132/3 - K1EL2	304 mm x 155 mm x 73 mm	Wall Mounted	45 lm / 6000k	1.5W	1
		Legrand Edgelight K132/3 - K1EL3	304 mm x 155 mm x 73 mm	Wall Mounted	45 lm / 6000k	1.5W	1

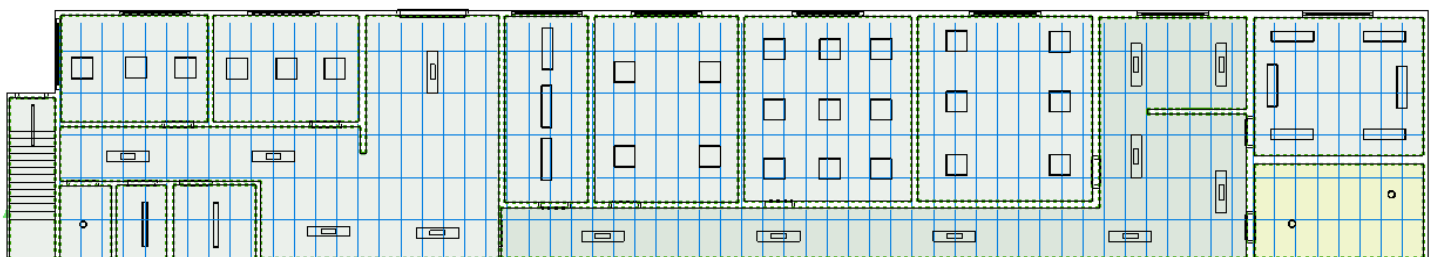
Table 24: Emergency Lighting Luminaire Schedule

Appendix G: Reflected Ceiling Plans

G1. Building A First Floor



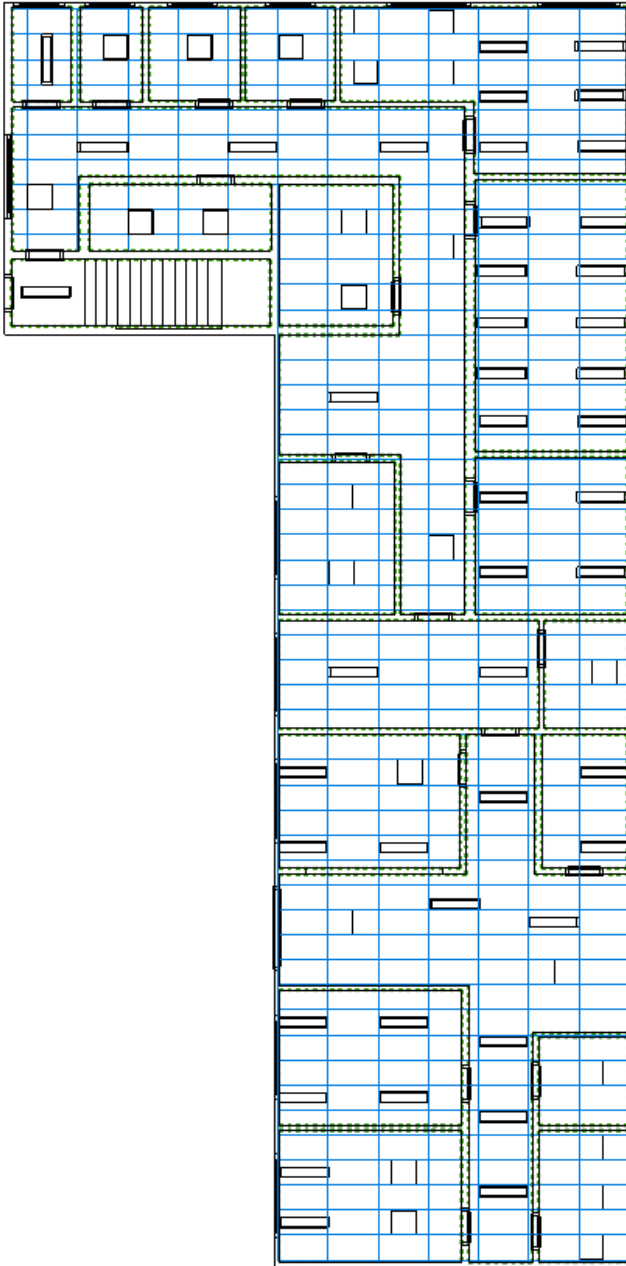
G2. Building A Ground Floor



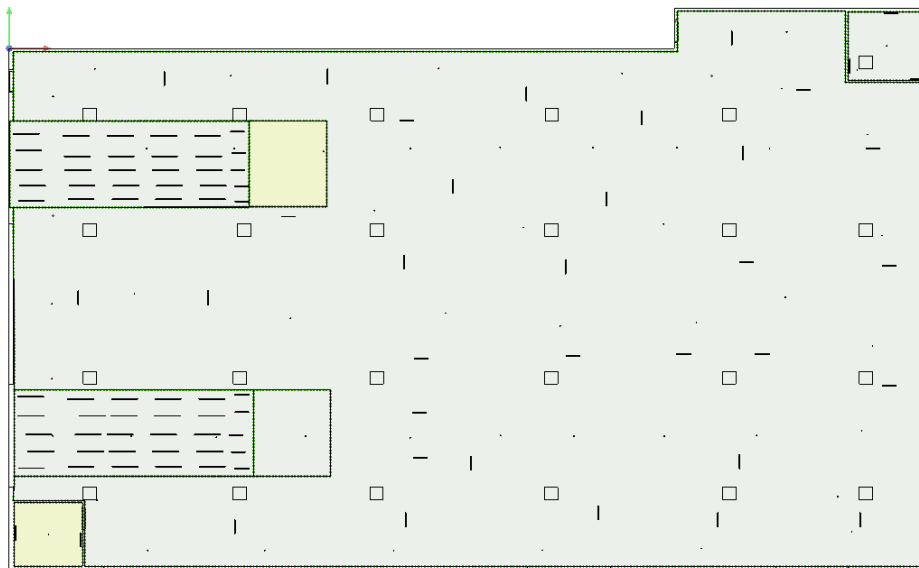
G3. Building B First Floor



G4. Building B Ground Floor



G5. Carpark



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